

SKILL-3

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2520030073 Sec-2

1. To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality. To Allocate Buffers Dynamically, Resize Arrays, Prevent BufferOverflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 10

int main()
{
    char *history[MAX];
    char command[100];
    int count = 0;
    while (1)
    {
        printf("Sushmitha$> ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = '\0';
        if (strcmp(command, "exit") == 0)
            break;
        if (strcmp(command, "history") == 0)
        {
            for (int i = 0; i < count; i++)
                printf("%d: %s\n", i + 1, history[i]);
            continue;
        }
        if (count < MAX)
        {
            history[count] = malloc(strlen(command) + 1);
            strcpy(history[count], command);
            count++;
        }
    }
    for (int i = 0; i < count; i++)
        free(history[i]);
    return 0;
}
```

```
bindu@Bindusree:~$ nano skill3a.c
bindu@Bindusree:~$ gcc skill3a.c -o skill3a
bindu@Bindusree:~$ ./skill3a
Bindu$> Hi
Bindu$> ls
Bindu$> pwd
Bindu$> history
1: Hi
2: ls
3: pwd
Bindu$> |
```

2. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output. To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
bindu@Bindusree: ~  
bindu@Bindusree:~$ nano skill3b.c  
bindu@Bindusree:~$ gcc skill3b.c -o skill3b  
bindu@Bindusree:~$ ./skill3b  
Enter command: ls 3  
Token: ls  
Token: 30  
bindu
```

```
GNU nano 8.7.1 sk:  
#include <stdio.h>  
#include <string.h>  
int main()  
{  
char input[100];  
char *token;  
printf("Enter command: ");  
fgets(input, sizeof(input), stdin);  
input[strcspn(input, "\n")] = '\0';  
if (strlen(input) == 0)  
{  
printf("Empty command\n");  
return 0;  
}  
token = strtok(input, " ");  
while (token != NULL)  
{  
printf("Token: %s\n", token);  
token = strtok(NULL, " ");  
}  
return 0;  
}
```